Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/31/2005
PI Name:	Jenkins, James T. Ph.D.		
Project Title:	Particle Segregation in Collisional Shearing Flows		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Ithaca	**State:**	NY
Zip Code:	14853	**Congressional District:**	22
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	05/28/2000	**End Date:**	12/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Rame, Enrique	**Contact Phone:**	216-433-2842
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2006, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Louge, Michel (Cornell University)		
Grant/Contract No.:	NCC3-797		
Performance Goal No.:	1.23		
Performance Goal Text:	Pursue ground-breaking basic research in fluid physics and transport phenomena to provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.		

Task Description:	In this work an axi-symmetric microgravity shear cell will be used to study granular flows. The cell incorporates two cylindrical boundaries that rotate in opposite directions. Both boundaries are fitted with appropriately designed bumps to achieve predetermined gradients in the fluctuation energy of the grains. These gradients produce gradients in the concentration of the two types of particles in the cell. The associated particle segregation is observed using digital video, image analysis, and sophisticated particle tracking algorithms. Two basic systems are to be examined: in one, the spheres are of equal size but differ in mass; in the other, they are of equal mass but differ in size. The observations will be compared to results of simulations and the predictions of theory to establish their respective limits and suggest possible improvements. A prototype shear cell has been flown on the KC135 microgravity aircraft and the results obtained from these experiments are being used in the design of the flight shear cell. Binary impact parameters needed as modeling information for the simulations and the theory are measured for collisions between two spheres and between a sphere and a boundary. The objective of this work is to investigate particle segregation in collisional flows of dry granular materials. This research has experimental, theoretical, and simulation components and focuses on the segregation of mixtures of binary sphere in situations in which spatial gradients in the granular fluctuation energy control the segregation. In systems that do not involve much agitation of the grains, several mechanisms that involve gravity have been identified as leading to segregation. In highly agitated flows there is a mechanism independent of gravity that is available to drive separation. This is associated with spatial gradients in the energy of the velocity fluctuations of the grains. Collisional interactions between and among different species of grains require that, in general, spatial gradients of concentration exist to balance spatial gradients of the particle fluctuation energy. In rapidly sheared or vibrated systems, gravity also influences mixtures of different size grains. Here, buoyant forces act to separate grains that differ in size and the local volume that they displace. In reduced gravity, buoyant convection is suppressed and attention can be focused on the simpler balance between gradients in concentration and the gradients in fluctuation energy. Reduced gravity also eliminates the possibility that a rapid flow will condense into a slower, denser flow dominated by enduring contacts rather than by collisions. Also, microgravity makes the visual observations possible by slowing down the segregation process, thus permitting one to employ moderate rates of shear. On Earth, the effects of gravity can be minimized by shearing so rapidly that the particle pressure overwhelms gravity. However, in this event, particles can be severely damaged which makes system characterization impossible, and separation takes place too rapidly for visual observation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The size segregation of flowing or shaken grains is a commonly observed phenomenon in industrial processes and in nature. In many industrial processes a homogeneous aggregate is desired; in these size segregation is undesirable. However, in the mining industry, segregation by size is exploited in some crushing operations. When observing natural grain deposits, grain segregation provides an indication of whether an aggregate of grains was deposited dry (larger grains above) or under water (larger grains below).
Task Progress:	Progress during 2004: Jenkins worked with Ricky Wildman of Loughboro University to describe the segregation profiles in binary mixtures of spheres that he observed above a vibrated bed in physical experiments. A simplified segregation theory derived by Birgir Arnarson was employed with an energy flux condition at a vibrating base developed by Mark Richman. The equations were solved in Matlab. The agreement between the observed and predicted profiles is excellent. David Yoon and Jenkins continued to work on applying kinetic theory to describe the segregation of a binary mixture disks in an annular shear cell. The predictions of the theory will be tested against data obtained in a shear cell installed on an air table at Rennes. David Yoon incorporated corrections to the kinetic theory for segregation of disks due to Alam, et al. into an existing linearized theory by Arnarson. David Yoon calculated the rates of collisional production of translational and rotational energy for frictional disks and used these to determine the partition of rotational and translational energy and an expression for the effective coefficient of restitution for slightly frictional disks. David Yoon and Jenkins developed the theory and completed a manuscript on a simplified mixture theory for disks, based on work done by Birgir Arnarson. These equations are also solved in Matlab. The theory will serve as the basis for the interpretation of experiments done on an air table at Rennes. Also, using the simplified mixture theory in the context of Matlab, David Yoon obtained segregation profiles for collisional flows of a binary mixture of spheres in a dense, inclined flow above an energetic base. David and Jenkins completed a manuscript on frictional dissipation for disks that should open the possibility of comparisons between the prediction of the kinetic theory and numerical simulations. Sovan Das and Jenkins completed a manuscript on steady, fully-developed, collisional flow in a vertical chute. David Yoon provided researchers at Loughboro University with a simplified mixture theory to permit them to compare the predictions of kinetic theory with the results of their experiments and numerical simulations on the segregation of binary mixtures in an axisymmetric container with a vibrating base.
Bibliography Type:	Description: (Last Updated: 11/27/2020)
Articles in Peer-reviewed Journals	Jenkins J, Johnson D, LaRagione L, Makse H. "Fluctuations and the effective moduli of an isotropic, random aggregate of identical, frictionless spheres." Journal of the Mechanics and Physics of Solids. 2005 Jan;53:197-225. https://doi.org/10.1016/j.jmps.2004.06.002 , Jan-2005
Books/Book Chapters	Jenkins JT. "The role of particle fluctuations in the mechanics of granular materials." in "Modelling of Cohesive-Frictional Materials." Ed. P.A. Vermeer, W. Ehlers, H. Herrmann, E. Ramm. Routledge publishing, 2004. p. 93-100., Oct-2004

Presentation	Jenkins, J.T.; Louge, M.Y.; Rame, E. "Granular Flow and In Situ Resource Utilization" Space Resources Roundtable 6, Golden, CO Oct-2004
Presentation	J.T. Jenkins, "Modeling the Mechanical Behavior of Granular Materials" Plenary Lecture, at the SIAM Conference on Mathematical Aspects of Materials Science, Los Angeles, May 23-27, 2004 May-2004
Presentation	J.T. Jenkins, "The Role of Fluctuations in the Mechanics of Granular Materials" Cohesive-Frictional Materials 2004, Stuttgart, September 2004 Sep-2004